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Python 3.13.12

Customer_Churn_Classification.ipynb

C:\> Users > lenovo > Downloads > Customer_Churn_Classification.ipynb > ML 4. Model Training with Cross-Validation > import numpy as np

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DATA TYPES AND MISSING VALUES

```
info_df = pd.DataFrame({
    'Data Type': df.dtypes,
    'Missing Values': df.isnull().sum(),
    'Missing %': (df.isnull().sum() / len(df) * 100).round(2),
    'Unique Values': df.nunique()
})
info_df
```

[24] ✓ 0.0s Python

	Data Type	Missing Values	Missing %	Unique Values
customerID	str	0	0.0	5880
gender	str	0	0.0	2
SeniorCitizen	int64	0	0.0	2
Partner	str	0	0.0	2
Dependents	str	0	0.0	2
tenure	int64	0	0.0	72
PhoneService	str	0	0.0	2
MultipleLines	str	0	0.0	3
InternetService	str	0	0.0	3
OnlineSecurity	str	0	0.0	3
OnlineBackup	str	0	0.0	3
DeviceProtection	str	0	0.0	3

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STATISTICAL SUMMARY - NUMERICAL FEATURES

df.describe()

[25] ✓ 0.0s Python

	SeniorCitizen	tenure	MonthlyCharges	TotalCharges
count	5880.000000	5880.000000	5880.000000	5880.000000
mean	0.500680	36.549150	70.157779	2566.813165
std	0.500042	20.909674	28.804615	1910.017743
min	0.000000	1.000000	20.000000	20.030000
25%	0.000000	18.000000	45.717500	1020.217500
50%	1.000000	37.000000	70.155000	2136.445000
75%	1.000000	55.000000	95.457500	3767.665000
max	1.000000	72.000000	119.990000	8589.600000

TARGET VARIABLE DISTRIBUTION (Churn)

Generate + Code + Markdown

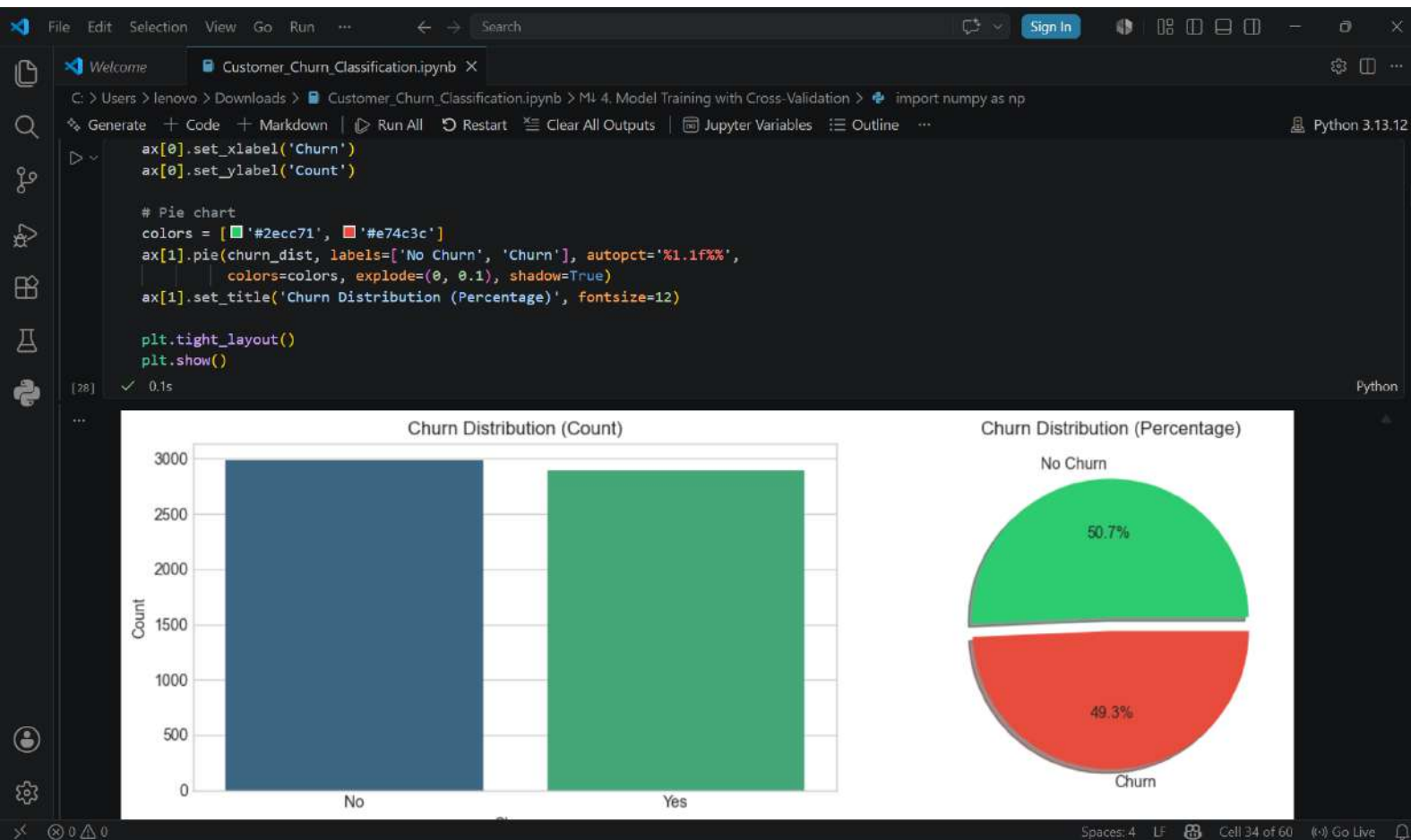
churn_dist = df['Churn'].value_counts()
churn_pct = df['Churn'].value_counts(normalize=True) * 10

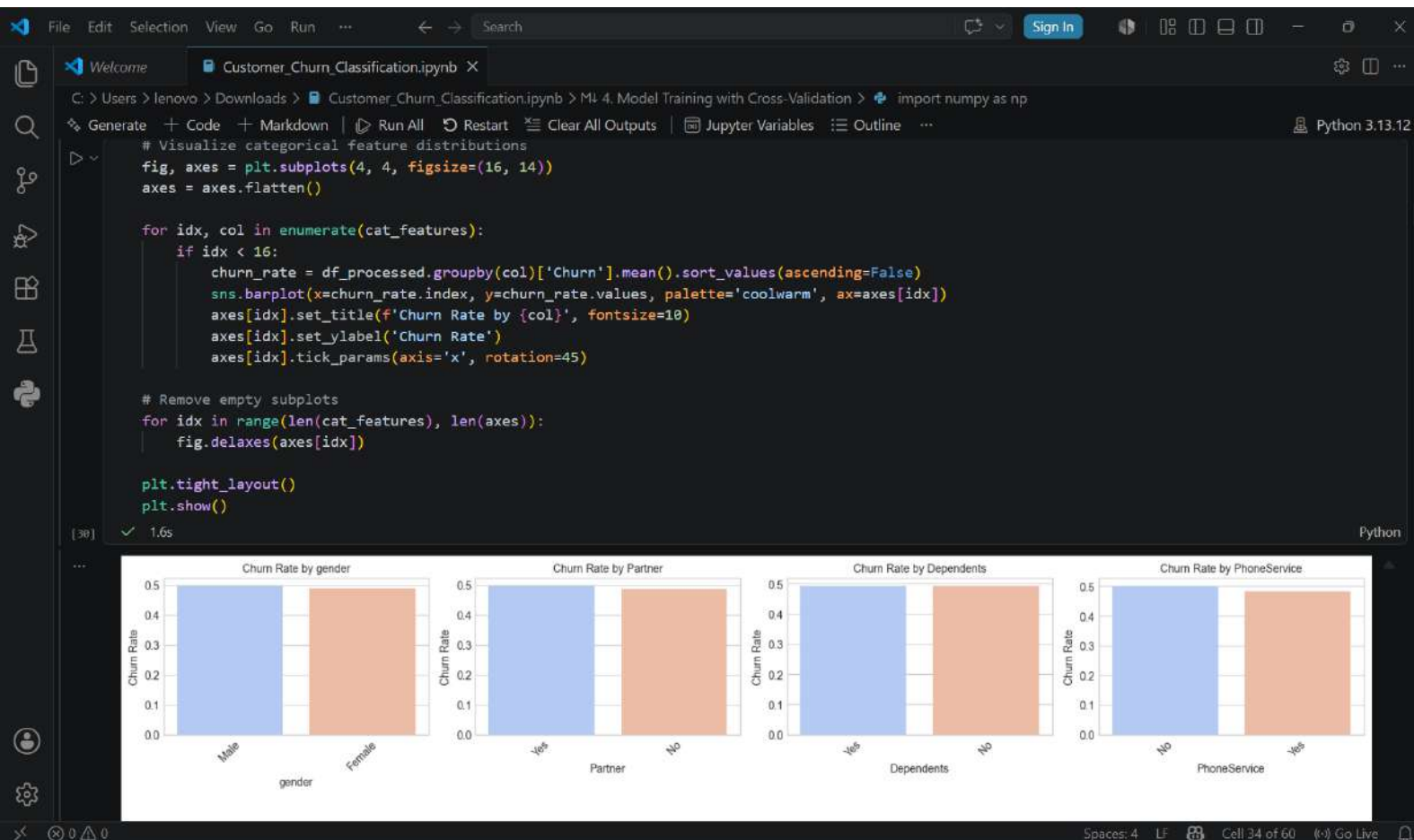
[26] ✓ 0.0s Python

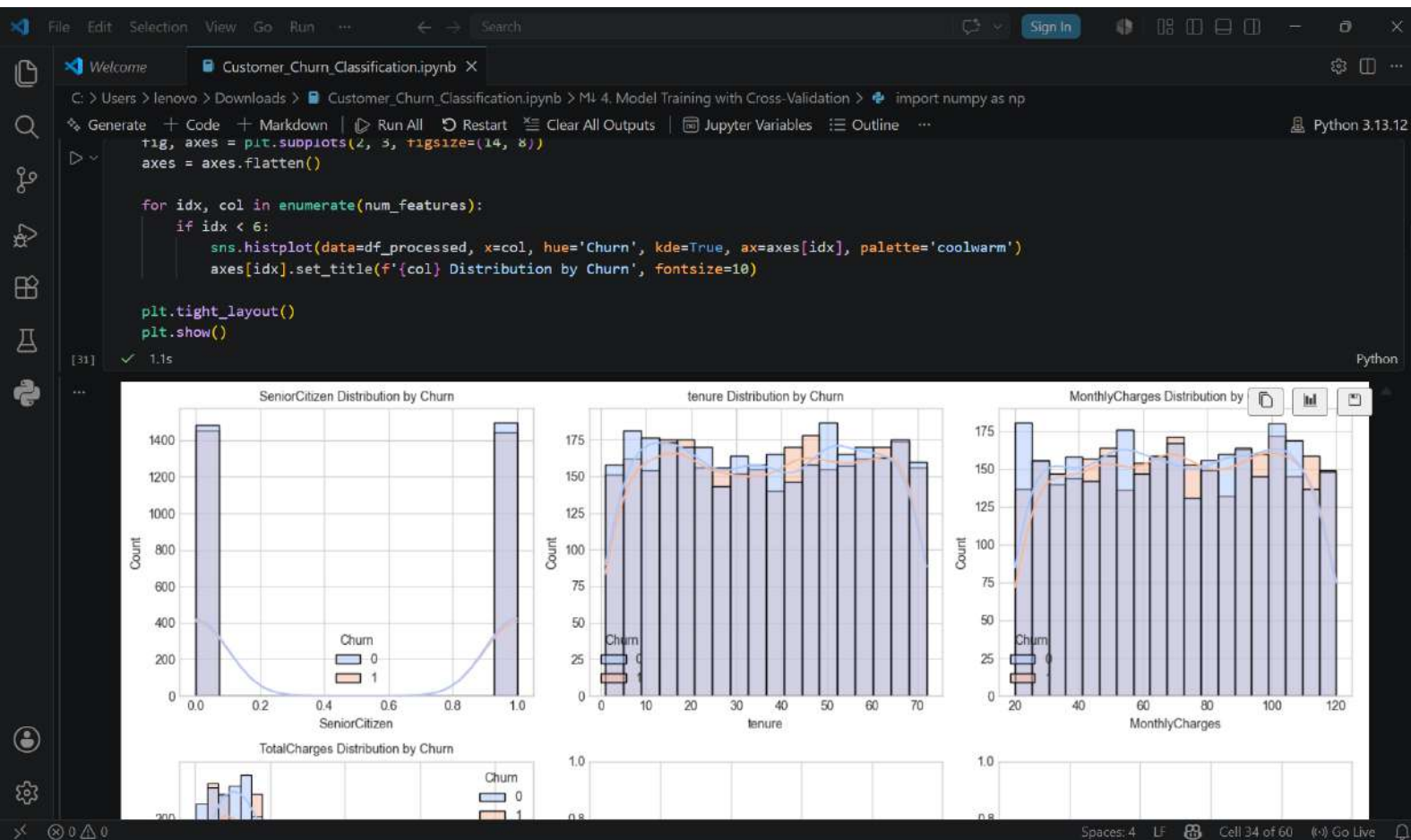
print(f"\nChurn Distribution:")

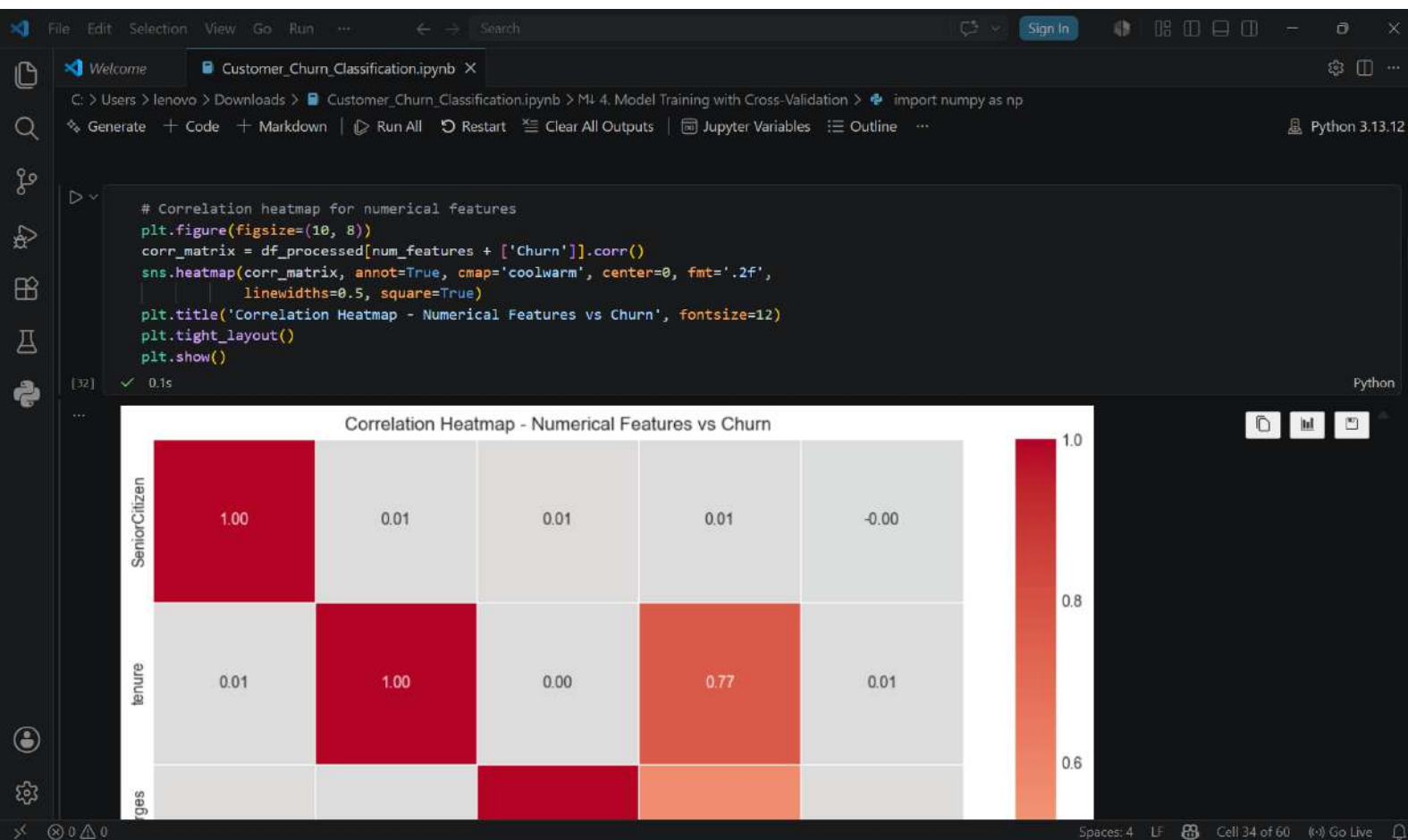
[27] ✓ 0.0s Python

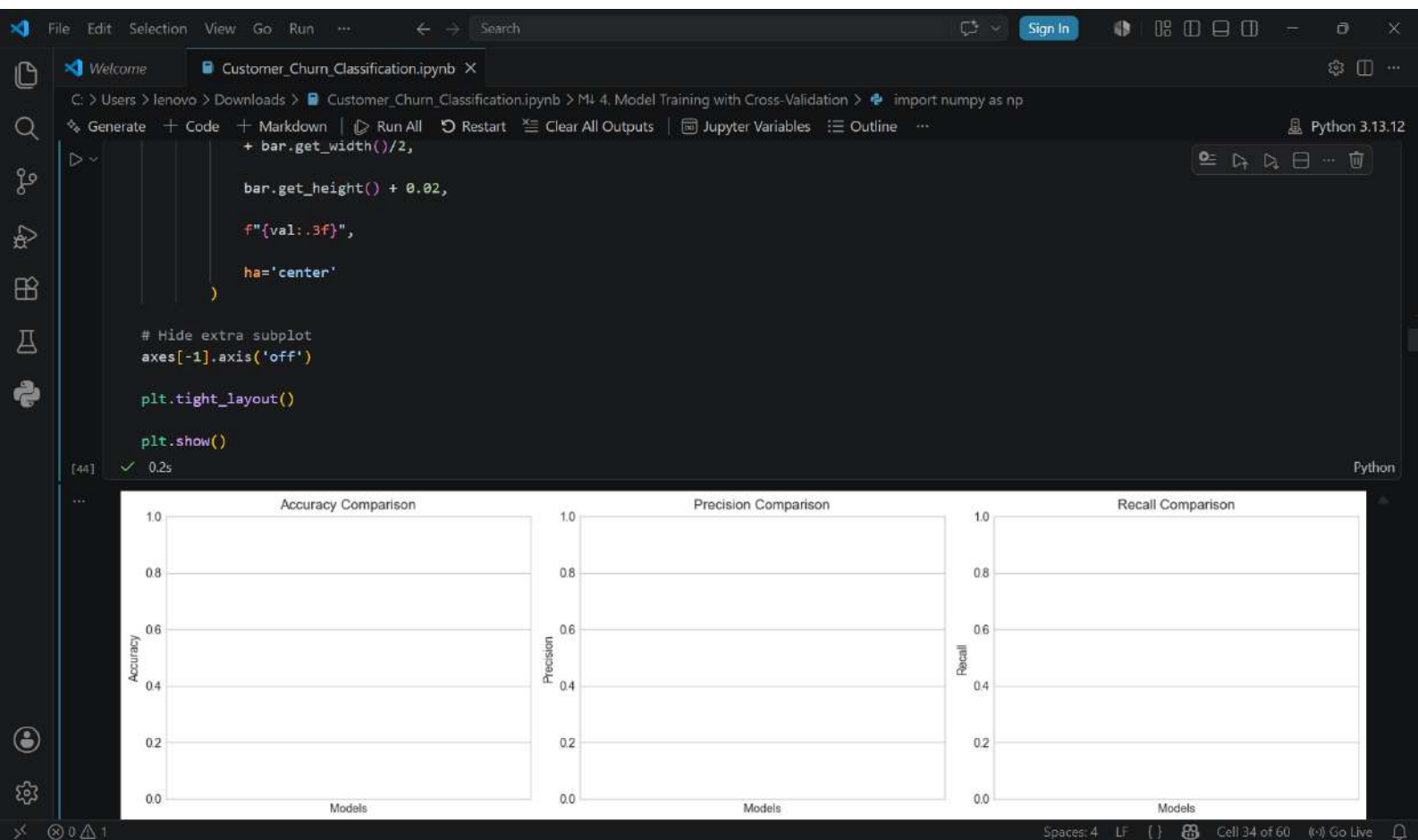
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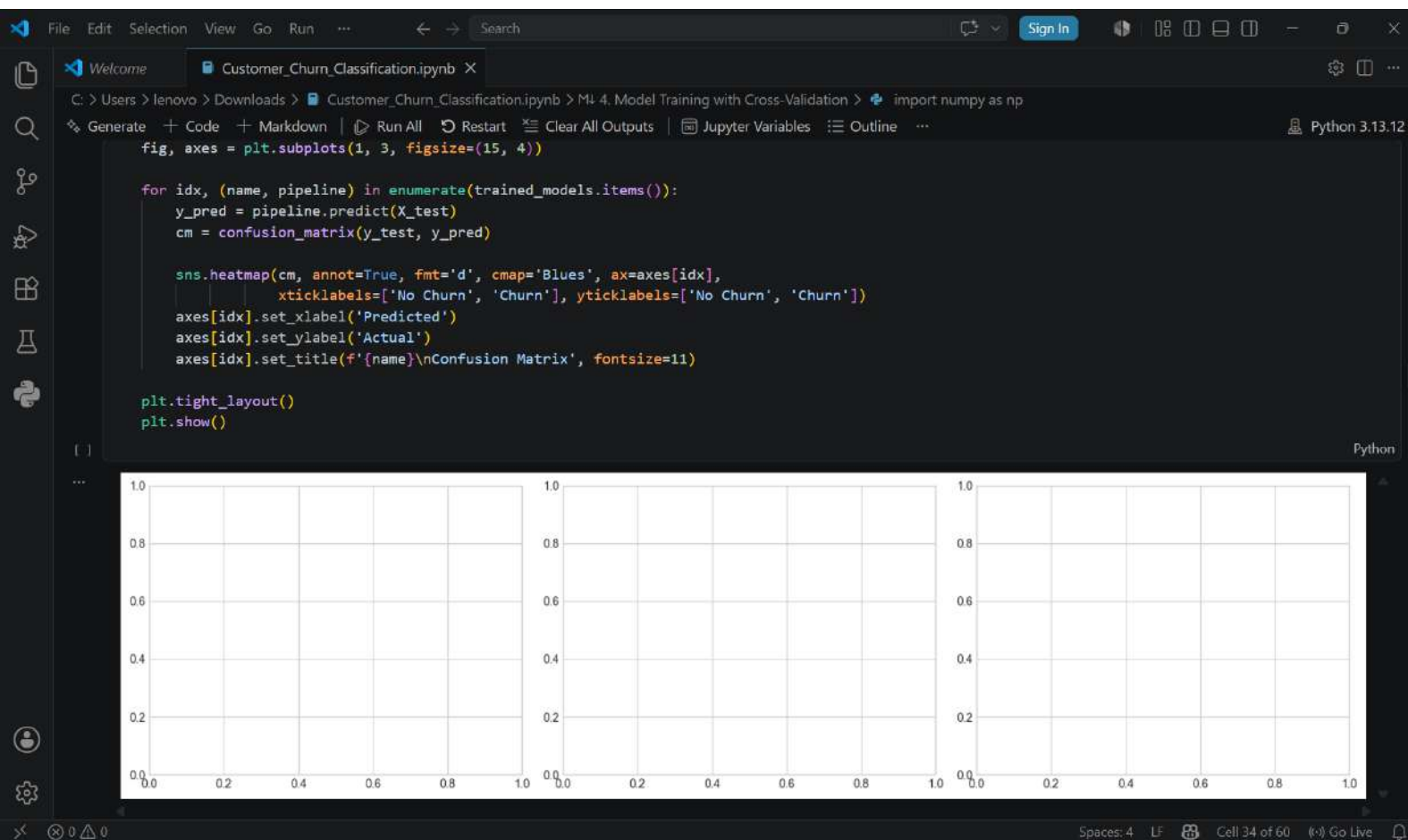


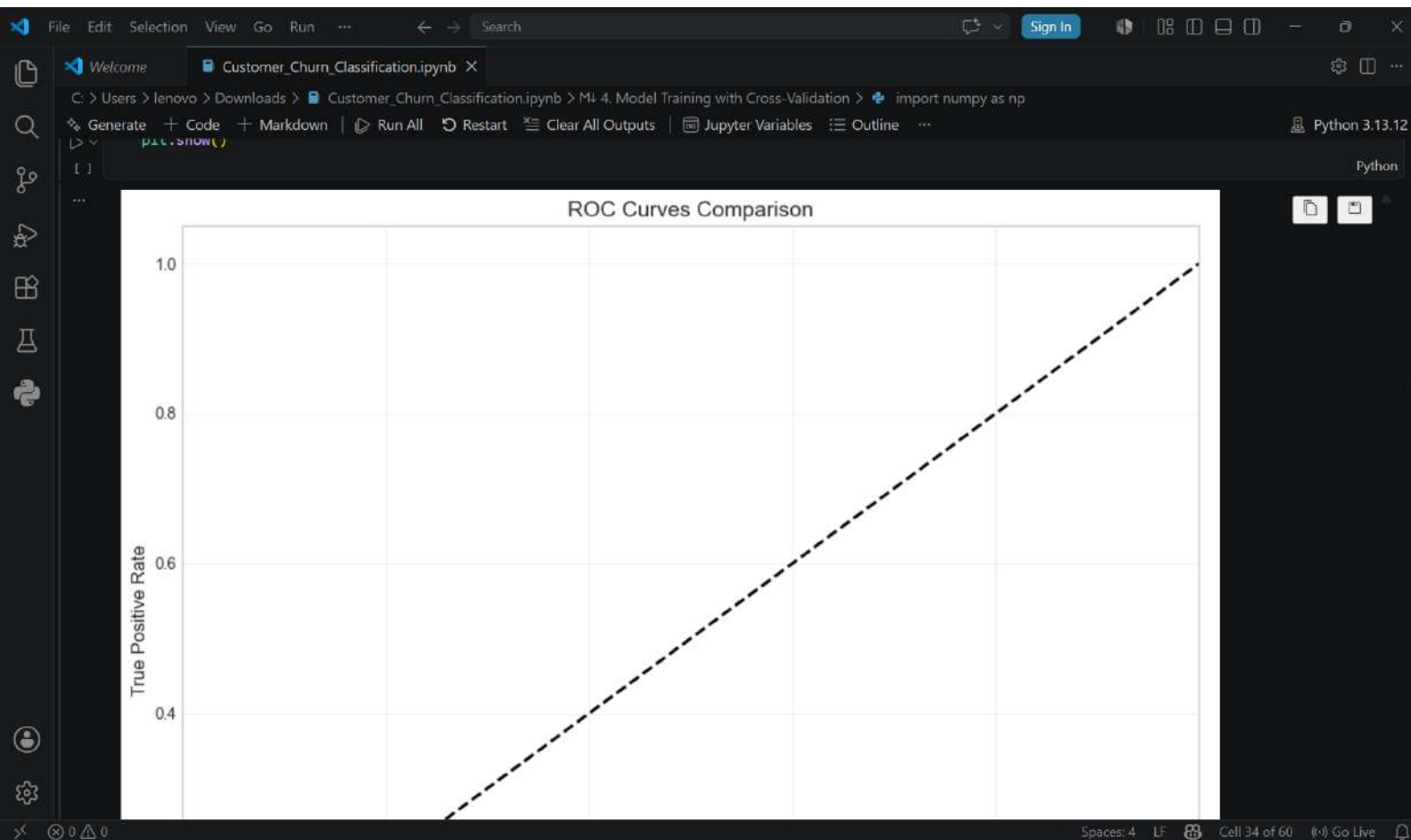


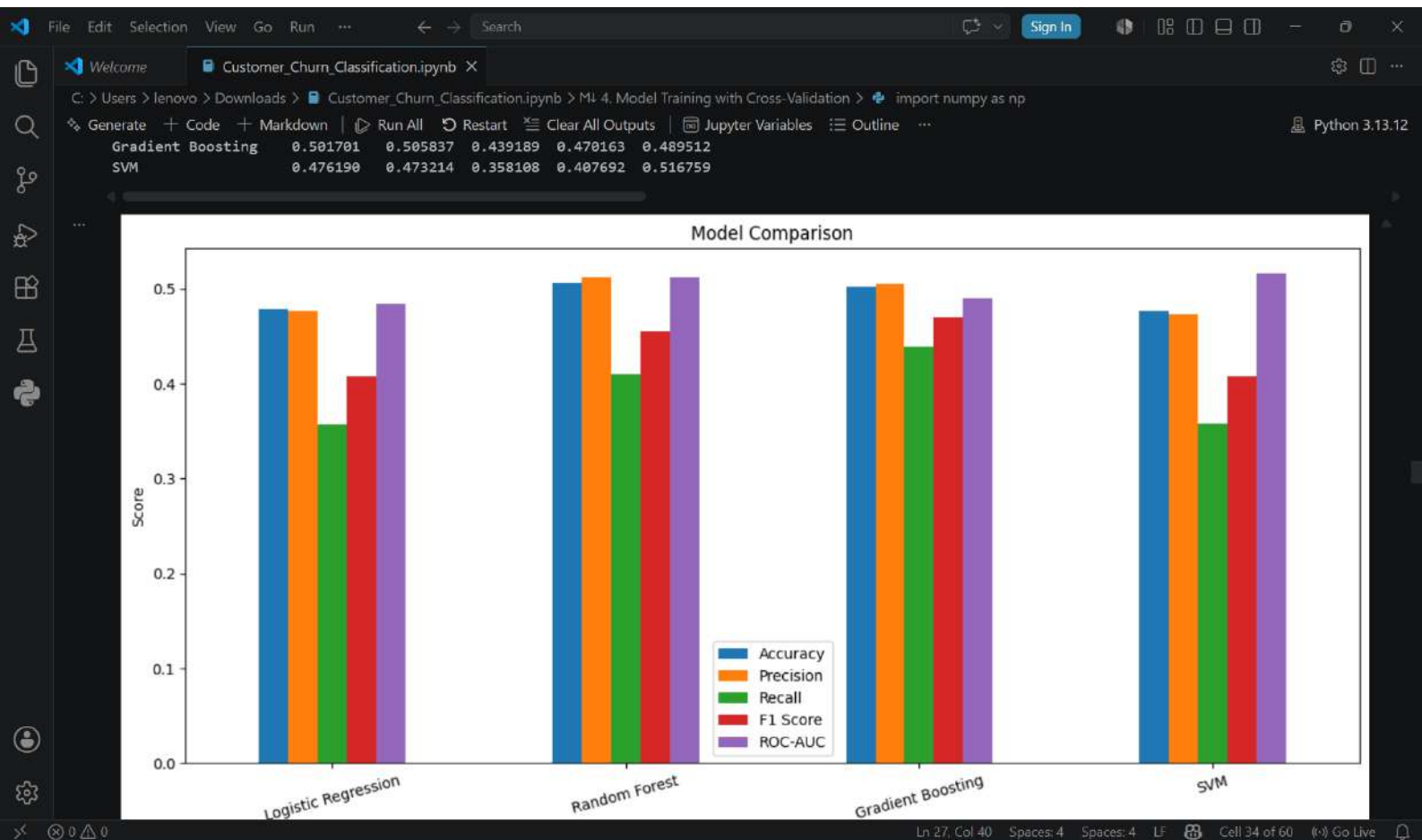












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```
print("=" * 70)

[ ]

...
=====
FINAL SUMMARY - CUSTOMER CHURN PREDICTION
=====

📊 DATASET OVERVIEW:
- Total Customers: 0
- Features: 19
- Churn Rate: 0.00%

📊 MODEL COMPARISON (Test Set):
-----
Model          Accuracy    Precision    Recall    F1          ROC-AUC
-----
Model results not available.
-----

🏆 BEST MODEL SUMMARY:
Best model details not available.

💡 KEY INSIGHTS:
1. Contract type is a strong predictor of churn
2. Month-to-month contracts have highest churn risk
3. Longer tenure reduces churn probability
4. Electronic check payment method is linked to higher churn
5. Fiber optic internet users often show higher churn rates

=====
PROJECT COMPLETED SUCCESSFULLY!
=====
```

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